onboardingManagement

Service Description

Service ID: *"onboardingManagement"*

Abstract
This document describes the `onboardingManagement` service, version 1.0.0, produced by the Onboarding support system of Arrowhead (documentation edition 5.3.0).

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1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

2 Overview

This document describes the `onboardingManagement` service, which provides operator approval and management of device enrollments. It is produced by the Onboarding system [1] in service version 1.0.0.

The rest of this document is organized as follows. In Section 3, we describe the abstract message operations provided by the service. In Section 4, we end the document by presenting the data types used by the mentioned operations. The concrete interface is specified in the HTTP/TLS/JSON IDD [2].

2.1 Significant Prior Art

Replaces the v4 shared-secret enrollment.

2.2 How This Service Is Meant to Be Used

An operator approves a device with `approve-device` (bootstrap-certificate fingerprint and the systems it may onboard), inspects enrollments, off-boards a device or removes the enrollment.

2.3 Important Delimitations and Dependencies

Needs an operator identity token bound to the operator certificate. An approval lists 1 to 32 system names for the device; reserved names (core, support and operator systems, configurable with `ONBOARDING_RESERVED_NAMES`) are refused.

3 Service Interface

This section describes the interfaces to the `onboardingManagement` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 4.

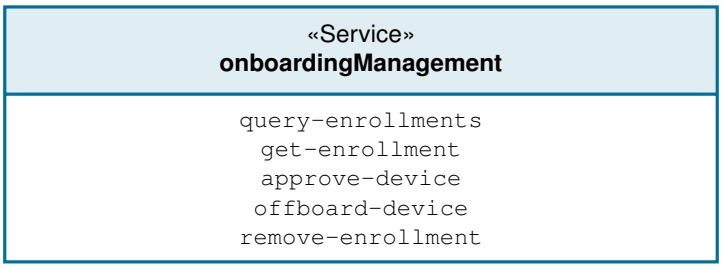


Figure 1: SysML-style block of the `onboardingManagement` service and its operations.

The following interface operations are available.

3.1 operation query-enrollments () : EnrollmentList

Returns all enrollments.

3.2 operation get-enrollment (Name) : EnrollmentView

Returns the enrollment of a device.

3.3 operation approve-device (EnrollmentRequest) : EnrollmentView

Approves a device: bootstrap-certificate fingerprint and the systems it may onboard.

3.4 operation offboard-device (Name) : OffboardingReport

Off-boards a device on behalf of the operator (as `onboarding.offboard-device`).

3.5 operation remove-enrollment (Name) : OffboardingReport

Off-boards the device and removes its enrollment.

4 Information Model

Here, all data objects that can be part of the `onboardingManagement` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 4.6, which are used to represent things like hashes and identifiers.

4.1 struct EnrollmentList

List of enrollments.

Field	Type	Description
entries	List<EnrollmentView>	Matching records of the current page.
count	Number	Number of enrollments.

4.2 struct EnrollmentRequest

Operator approval of a device.

Field	Type	Description
bootstrapFingerprint	String	SHA-256 fingerprint of the approved bootstrap certificate.
systems	List<String>	Systems the device may onboard.

4.3 struct EnrollmentView

An enrollment (approved device).

Field	Type	Description
deviceName	Name	Name of the device hosting the system.
bootstrapFingerprint	String	SHA-256 fingerprint of the approved bootstrap certificate.
approvedSystems	List<String>	Systems the operator approved for the device.
state	String	Enrollment state: `APPROVED`, `DEVICE_REGISTERED` or `OFFBOARDED`.
deviceCertificateId	Number	CertificateAuthority record of the device certificate.
deviceCertificateExpiresAt	DateTime	Expiry of the device certificate.
systems	List<SystemEnrollmentView>	Systems onboarded for the device.
approvedBy	String	Operator who approved the device.
approvedAt	DateTime	Time of the approval.
updatedAt	DateTime	UTC time of the last change.
lastMessage	String	Last status message of the enrollment.

4.4 struct OffboardingReport

Outcome of an off-boarding.

Field	Type	Description
deviceName	Name	The off-boarded device.
completed	List<String>	Off-boarding steps that succeeded.
failed	List<String>	Off-boarding steps that failed.

4.5 struct SystemEnrollmentView

A system onboarded for a device.

Field	Type	Description
systemName	Name	Name of the system, PascalCase (e.g. `TemperatureSensor`).
certificateId	Number	CertificateAuthority record of the system certificate.
certificateExpiresAt	DateTime	Expiry of the system certificate.
issuedAt	DateTime	Time the certificate was issued.

4.6 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

Type	Description
DateTime	Pinpoints a specific moment in time (UTC).
List	An *array* of a known number of items, each having type A.
Name	A string identifier that is intended to be both human and machine-readable.
Number	An integer number.
String	A chain of characters.

5 References

[1] J. Delsing, “Onboarding – System Description,” Eclipse Arrowhead, SysD 5.3.0, 2026, ah5-Onboarding-java-spring/documentation/v5.3.0/Onboarding-v5.3.0-SysD.pdf.

[2] ——, “onboardingManagement HTTP/TLS/JSON – Interface Design Description (Onboarding),” Eclipse Arrowhead, IDD 5.3.0, 2026, onboarding-v5.3.0-onboardingManagement-HTTP-IDD.pdf.

6 Revision History

6.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): approval limits (1 to 32 systems, reserved names) added.	Claude Opus 5.5 (draft)
3	2026-11-09	5.3.0	New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained.	Documentation maintenance

6.2 Quality Assurance

No.	Date	Version	Approved by
1		5.3.0	